Rules for differentiation

1 Differentiate the following:

a $y = 3x^2$ b $y = 6x^4$ c $y = -4x^3$ d $y = 5x^5$

e $y = -\frac{x^3}{5}$ f $y = \frac{3x^6}{2}$ g $y = \frac{5x^4}{12}$

2 Differentiate the following, expressing your answers in positive index form.

a $y = \frac{14}{x\sqrt{x}}$ b $y = \frac{1}{4x^3}$

3 Differentiate the following:

a $y = x - 9$ b $y = 2x + 9$

c $y = \frac{2x}{9} + 7$ d $y = x^2 - x + 8$

e $y = x^2 - 8x - 6$ f $y = x^4 + x^5$

g $y = x^5 - x^4 + 3$ h $y = \frac{1}{2}x^5 + \frac{1}{5}x^8$

i $y = \frac{x^8}{8} + \frac{x^5}{5} - 3x$

4 For each of the following functions:

i Rewrite the function in expanded form. ii Differentiate the function.

a $y = \frac{4}{9}(-4x - 8)$ b $y = (6x + 5)(x + 3)$

c $y = 2x^2(7x + 2)$ d $y = x(3x + 4)(5x + 6)$

e $y = (x + 4)^2$ f $y = 5(x - 3)^2$

g $y = (8x - 4)^2$

5 Differentiate the following functions:

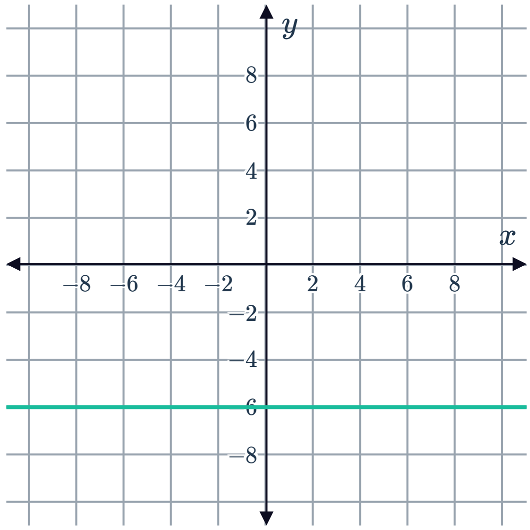
a $y = (3x + 2)(7x + 6)$ b $y = (x + 8)(x - 7) + 5$

Gradients

6 Find $f'(2)$ if $f'(x) = 4x^3 - 3x^2 + 4x - 6$.

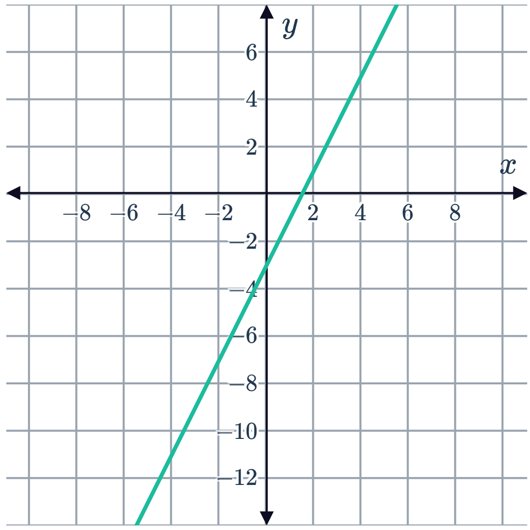
7 Consider the graph of $f(x) = -6$ shown:

Find $f'(4)$.



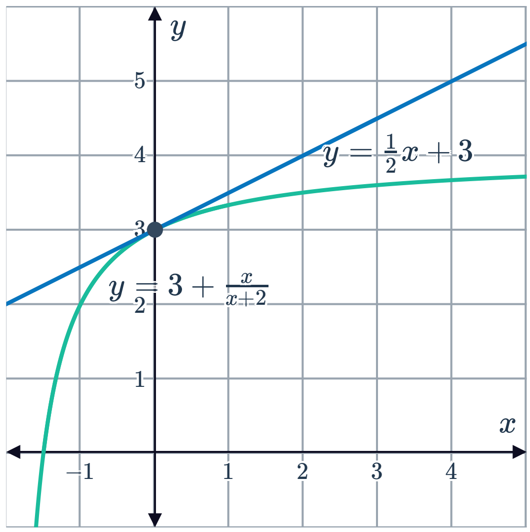
8 Consider the graph of $f(x) = 2x - 3$:

Find $f'(-4)$.

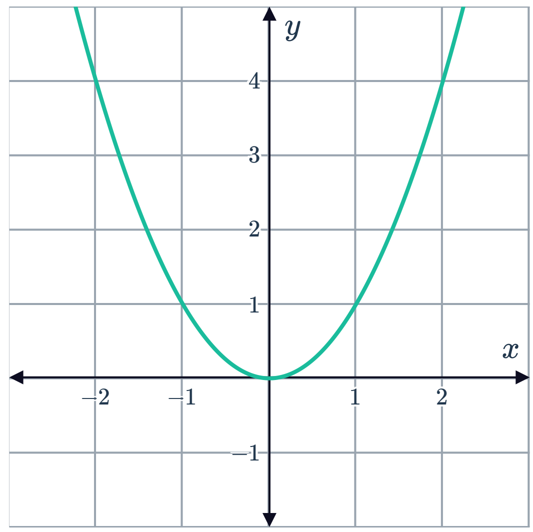


9 The tangent to the curve $y = 3 + \dfrac{x}{x+2}$ at the point $(0, 3)$ has the equation $y = \dfrac{1}{2}x + 3$:

Find $f'(0)$.



10 Consider the graph of the function $f(x) = x^2$:



a How many points on the graph of $f(x) = x^2$ have a gradient of 2?

b Find the x-coordinate of the point at which $f(x) = x^2$ has a gradient of 2.

11 Find the gradient of $f(x) = x^4 + 7x$ at the point $(2, 30)$:

12 Consider the function $f(x) = 6x^2 + 5x + 2$.

a Find $f'(x)$.

b Find $f'(2)$.

c Find the x-coordinate of the point at which $f'(x) = 41$.

13 Consider the function $f(x) = x^3 - 4x$.

a Find $f'(x)$.

b Find $f'(4)$.

c Find $f'(-4)$.

d Find the x-coordinates of the points at which $f'(x) = 71$.

14 Consider the function $y = 2x^2 - 8x + 5$.

a Find $\frac{dy}{dx}$.

b Hence, find the value of x at which the gradient is 0.

15 Find the x-coordinates of the points at which $f(x) = -3x^3$ has a gradient of -81.